

Written Report 1

Group Members: Aimar Mardones, Alejandro García, Oier Abrisqueta, Jon Sacha Díez

[LINK TO THE DATASET](#)

Section 1 - Descriptive Analysis

Motivation

In today's digital era, everybody has been given the opportunity to create, this has led to an extreme competitiveness, thus society is more critical with these creations. In this context, our team has chosen to study what drives a news article to become “viral”, specifically we are studying Online News Popularity.

We are interested in learning which characteristics have a real relation with the popularity of the article. By applying statistics models, our purpose is to uncover the underlying patterns that separate a regular article from a viral one.

Dataset Selection

In order to study this phenomenon, we have reached a dataset that summarizes a heterogeneous set of features about articles published by Mashable.

Mashable is a digital media platform that publishes news and content mainly related to tech, science, creation, social media, entertainment and daily life.

We have selected the Online News Popularity dataset from the UC Irvine Machine Learning Repository. This dataset contains a broad set of diverse features of articles and a large volume of observations, which let us perform the optimal techniques to obtain the desired results.

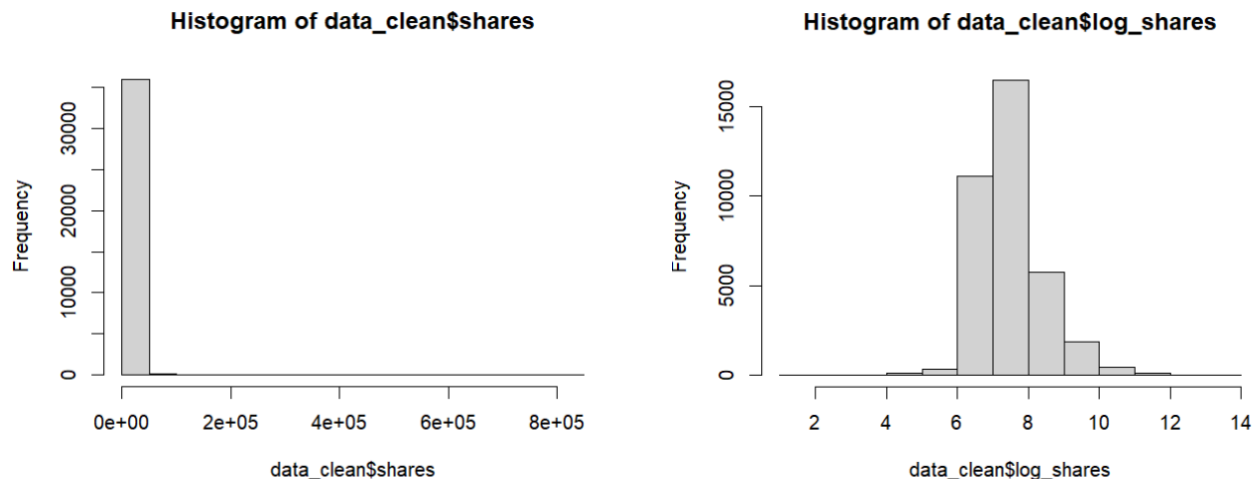
Characteristics of the Dataset

The dataset accomplishes all the requirements:

- Sample size: the dataset has 39797 observations.
- Binary Variables: it includes multiple binary variables, such as **is_weekend** or **data_channel_is_tech**
- Continuous variables: there are above 8 continuous variables.

Variable explanation and data cleaning

We evaluated the distribution using hist and density functions. We observed that the distribution was right-skewed as evidenced by the mean being substantially higher than the median.



To address this skewness and satisfy the normality assumption, we applied a logarithmic transformation to the target variable creating the variable **log_shares**. By applying this change, we realised that the distribution now is a little bit left-skewed which is a result of the high number of observations with a very low number of shares. This transformation is crucial for creating correct and consistent linear and logistic models.

The original dataset contained several binary variables for the day of the week (`weekday_is_monday` , `weekday_is_tuesday`...). To improve the interpretation of our future models, we have converted all these binary variables into a factor named **day_of_week**.

According to the dataset documentation, `n_unique_tokens` and `n_non_stop_unique_tokens` told nearly the same information so we decided to study its correlation coefficients. A coefficient of 0.88 was uncovered so we decided to study the correlation of both variables with the target variables and as **n_unique_tokens** had a lower correlation coefficient, we decided to eliminate it.

We evaluated the possibility that the variables starting with `kw` such as `kw_min_max`, `kw_avg_max` or `kw_max_min` carried redundant information. To address this question we studied a correlation matrix between these variables and extracted the pairs with > 0.8 correlation coefficients, which we considered it was a quite high correlation. We discovered that **kw_avg_min** and **kw_max_min** had a 0.94 correlation coefficient in the matrix. Then, as before, we compared the correlation coefficients of these two variables with the target variables, shares, and we eliminated the one with less correlation with the target that was **kw_max_min**.

We evaluated the count variables, which represent the frequency of elements (words, images, links) within each article. To ensure model robustness, we filtered the 99th

percentile for **n_tokens_content**, **num_hrefs**, **num_imgs**, **num_videos**, and internal references.

Regarding NLP metrics, we excluded **n_non_stop_words** due to its near-zero variance. With the mean (0.996), 3rd quartile (1.000), and 95th percentile (1.000) virtually identical, the variable lacks the necessary spread to explain changes in the target variable (shares). Statistically, a near-constant predictor provides no information and may lead to instability in the regression matrix. For **n_non_stop_unique_tokens**, we removed records below 0.25 to eliminate non-informative or repetitive content, while **num_keywords** was kept as it showed no significant outliers.

Articles that had no words had as well all columns Na so that it did not make sense keeping them. Though probably they were videos or other type of content, as they did not have real values on them they were not worth analyzing.

We drop as well the column **kw_min_min**, owing to the fact that for 19 460 articles it had a -1 value that worked as an “unknown” and it did not make sense deleting all those articles because they represent the 55% of our data.

Following the initial data cleaning, a rigorous feature selection process was implemented to address multicollinearity. Variables were first grouped by functional characteristics to identify thematic redundancies. Within these groups, we conducted a pairwise correlation analysis, extracting variables with a correlation coefficient exceeding **0.8**. To determine which features to retain, we employed a relevance-based filtering strategy: in every instance of high inter-correlation, the variable demonstrating the strongest absolute correlation with the target, **log_shares**, was preserved while the other part of the pair was discarded. This systematic approach ensures that the final dataset retains maximum predictive signal while minimizing the noise.

Finally, following the last step of analyzing the correlation between predictors, we analyzed the correlation of LDA variables and did not find any correlation superior to 0.8. The variables **LDA_00** to **LDA_04** represent the latent thematic distribution of each article, obtained using the Artificial Intelligence algorithm 'Latent Dirichlet Allocation' (NLP). These values are continuous proportions (summing to 1.0) and indicate the relative weight of each of the 5 major hidden themes discovered by the AI within the article's text. In our model, this acts as the 'thematic DNA' or context of the article.

Section 2 - Regression models

1. Linear regression models

1.1. Linear regression model based on content of the article

Objective

The idea behind this linear regression model is to study if there is a significant relationship between linguistic characteristics such as **polarity** or **subjectivity** and the resulting engagement measured by **log_shares**.

Initial Steps

To test this hypothetical relationship, we used a specific subset of predictor variables focusing on article content, article tokens, polarity, subjectivity and word rates. Then, to check multicollinearity we built a correlation matrix between all the predictors to eliminate highly correlated variables (> 0.8). As in data cleaning, the filtering strategy used was to keep the predictor of the pair that is more correlated with the response, log_shares. This step was critical to prevent the inclusion of redundant data that would otherwise add noise to the model and inflate standard errors.

Once this step was completed, we moved on data splitting in which data was partitioned into:

- **Training Set (70 %)** → Used for parameter estimation and model fitting.
- **Test Set (30 %)** → Reserved for later performance evaluation.

We ensured that the split was performed randomly to maintain a representative distribution of variables across both sets, providing a robust benchmark for our final R squared and Mean Squared Error (MSE) metrics.

Models

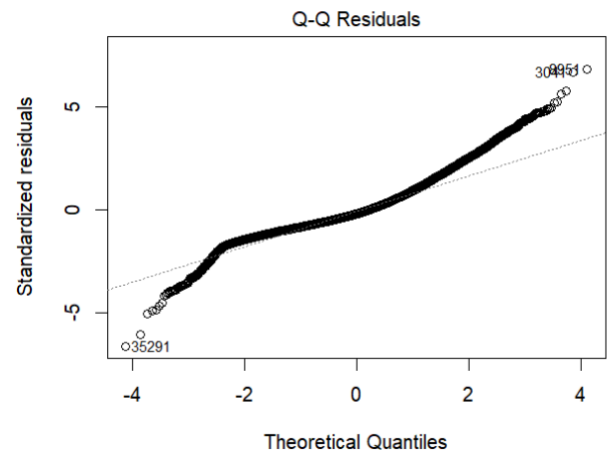
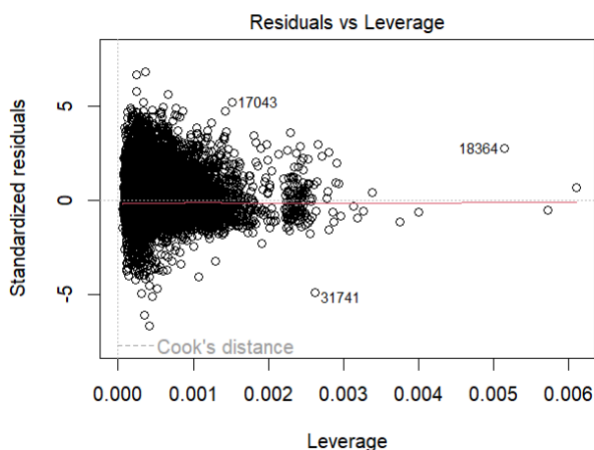
A complete linear regression model was built with all remaining variables and the F-test was performed under the null hypothesis “none of the predictors tell relevant information”. As the **p-value was very small ($< 2.2e-16$)**, we concluded that there is a significant linear relationship between the response variable and, at least, one of the predictors.

After this, we decided to follow a backward elimination approach to keep only relevant variables into the model. This iterative process began with the full set of pre-selected content predictors, systematically removing the least significant variables one at a time. This process followed a dual-testing criterion at each iteration:

- **Individual significance (T-test)** → We examined the p-values associated with each predictor's T-statistic. The variable with the highest p-value (providing it exceeds the threshold value $\alpha = 0.05$) was identified as a candidate for removal.
- **Model Comparison (Anova)** → Following each potential elimination, an ANOVA is performed to compare the reduced model against the previous one. This ensured that the removal of the variable did not result in a significant loss of the model efficacy.

Model Diagnosis

Using the final model, we calculated 95% confidence intervals and compared their endpoints with the fitted values of the target variable (log_shares) to assess the model's prediction errors. Furthermore, we performed a model diagnosis using various diagnostic plots, for instance, q-q quantile plot or residuals vs fitted plot.



The Q-Q Residuals plot explains that our model has normality issues:

- **S-Shape** → Instead of the points following the dashed straight line, they deviate significantly at both ends.
- **Heavy tails** → The points are much higher than the line in both left and right tails. This indicates that our data has more outliers or extreme values than the ones a normal distribution would expect.
- **Verdict** → The model is underestimating the probability of extreme values of data. A standard linear regression is based on the assumption that errors are normally distributed. However, this plot suggests that there is a non-normality of errors as evidenced in the heavy and deviated tails.

The Residuals vs Leverage plot shows that our model does not have high influential points:

- **Bulk of points** → The vast majority of data points are located between 0 and 0.003 of leverage and between -5 and 5 of Standardized residuals.
- **Outliers** → There are some points with higher leverage but never increasing 0.007.
- **Verdict** → This plot suggests that, although there are some outliers in the prediction, these points are not influential at all and are not pushing the line towards them.

The analysis confirms that article content, specifically structural features, subjectivity, and sentiment holds a statistically significant linear relationship with the popularity of online news evidenced by the low p-value. Through confidence-interval studying, we identified that global

subjectivity is the strongest positive driver of shares, while the length of the title and content tends to have a slight negative impact. Nevertheless, the R-squared statistic of 0.025 uncovers that our model only explains the 2.5% of the total variance in log_shares. While the extremely low p-value confirms that the relationship between article content and popularity is statistically relevant and not explained due to random elements, the low R2 highlights the complexity of social media virality that does not follow a constant relationship. In the context of the Online News Popularity dataset, this suggests that while the selected features like subjectivity and title length provide clear evidence of predictive success, the remaining 92% of the variability is actually driven by factors outside the scope of this study.

1.2. Linear regression model based on editorial strategy and publication timing

Objective

The objective of this model is to study the influence of editorial strategy and publication timing in the shares of an article. To do so, we make a model which has the weekday publication, the data channel and the LDA as predictors.

Initial Steps

These three variables are all categorical variables, that's why we will need to have k-1 predictors from each variable group to avoid the dummy variable trap. In the case of the weekday publication we have eliminated Sunday, from the data channel we haven't eliminated any of the variables because there are observations with all zeros in them (the ones which are segmented as other channels) and from LDA we have eliminated the LDA_04. Now the results of the estimated values of each variable within the group of the same category will represent their comparisons to the base one, that is the one eliminated (weekday_is_sunday, LDA_04 and other data channels).

Once we have eliminated the dummy variables we split the data into:

- **Training Set (70 %)** → Used for parameter estimation and model fitting
- **Test Set (30 %)** → Reserved for later performance evaluation

Model

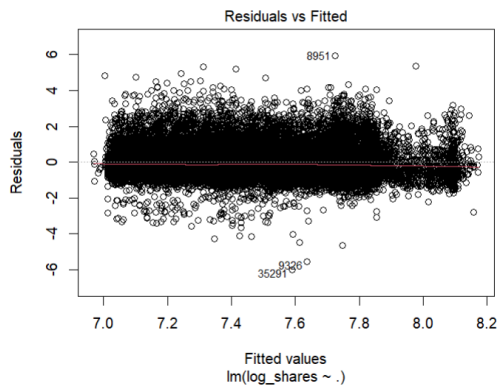
With the training set we built a multiple linear regression model and the F-test was performed under the null hypothesis "none of the predictors tell relevant information". As the **p-value was very small ($< 2.2e-16$)**, we concluded that there is a significant linear relationship between the response variable and, at least, one of the predictors.

R2 value is 0.07708 which means that the model represents 7,7% of the variance of the variability of an article.

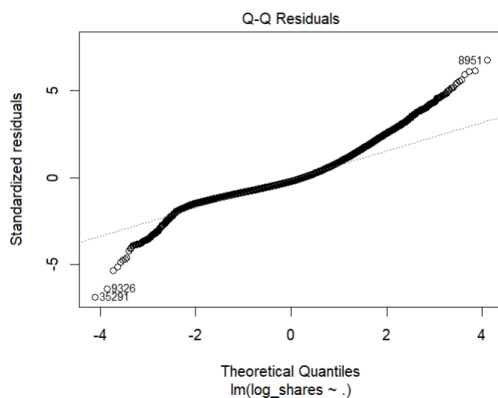
The model also shows that sunday and saturdays are the best days to publish the articles. Sunday is the base predictor between weekdays category and all the remaining variables have a negative estimated value except Saturdays that are similar to Sundays. In the data channel category the most viral articles are the ones that aren't segmented, being followed by the social media articles. Finally, the articles that IA category with more LDA_00 and LDA_03 content are the ones which work better.

Model Diagnosis

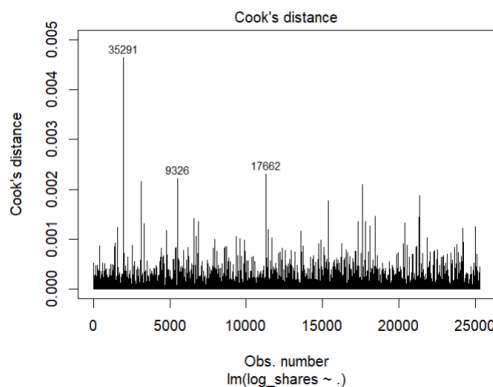
Once we have our final model and we understand how and if the model works, we are going to check if the model is adequate, checking if it satisfies all the criterias.



First we will use the residual plot. In this graph we can see that the residuals should be randomly scattered around zero, without any clear pattern, which means that it satisfies the non-linearity criteria. As there isn't any clear pattern we can assume that correlation of errors is also satisfied, a problem usually ligated to observations linked to time. Non-constant variance errors are also satisfied because as the graph shows our model presents homoscedasticity, with no funnel-like shape.



This is a quantile-quantile plot(Q-Q plot) a graph used to check if the errors follow a normal distribution. It shows an S-shape graph that reveals the errors don't follow a normal distribution, however it is normal when speaking about real data and more specifically in virality of articles.



Finally, we check the presence of high-leverage points using the Cook's Distance plot, where we can see the presence of three high leverage, high influential points.

2. Logistic Regression models

2.1 Logistic Analysis about the influence of media and keywords on virality

Objective

The aim of this analysis is to identify whether the virality of an article depends on the media and keywords it uses or not. For this purpose, we have developed 4 logistic regression models, evolving from a superficial baseline (Model A) to a calibrated, context aware and non linear predictive model (Model B with Splines).

Initial Steps

The initial steps taken have been to identify whether an article is viral or not. We have defined the articles whose shares have reached the top 25% as viral. The next step has been to superficially identify the trends between virality and the variables we were interested in (number of images, videos and kw_avg_avg) through graphic visualization. We have as well divided the previously cleaned dataframe into a training and testing dataset, allocating the 70% of data for training and the 30% for testing. For this division we have applied the Stratified Sampling technique to ensure proportional representation of viral and non-viral articles.

Models

As previously mentioned, for this analysis 4 models have been employed:

- model_A: num_imgs, num_videos and kw_avg_avg are the only predictors. Even though the main objective is to analyze the relationship between these exact predictors and virality, the lack of control variables that have also an influence on the shares of an article was leading into an inaccurate result as we will observe later.
- model_B: besides the original predictors this model takes control variables into account. We have considered the following control variables: "n_tokens_content", "n_non_stop_unique_tokens", "num_hrefs", "num_self_hrefs", "average_token_length", "self_reference_avg_shares", "global_subjectivity", "global_sentiment_polarity", "rate_positive_words", "rate_negative_words", "day_of_week", "data_channel_is_lifestyle", "data_channel_is_entertainment", "data_channel_is_bus", "data_channel_is_socmed", "data_channel_is_tech" and "data_channel_is_world".

Though initially our plan was to study only these two models, their evaluation has led us to the realization that they were poorly calibrated. As human behaviour rarely scales in a straight line, with the aim of reducing linearity, Natural Cubic Splines were applied to continuous variables of both models. So the other 2 logistic models are:

- model_A_nonlinearity
- model_B_nonlinearity

Model Evaluation

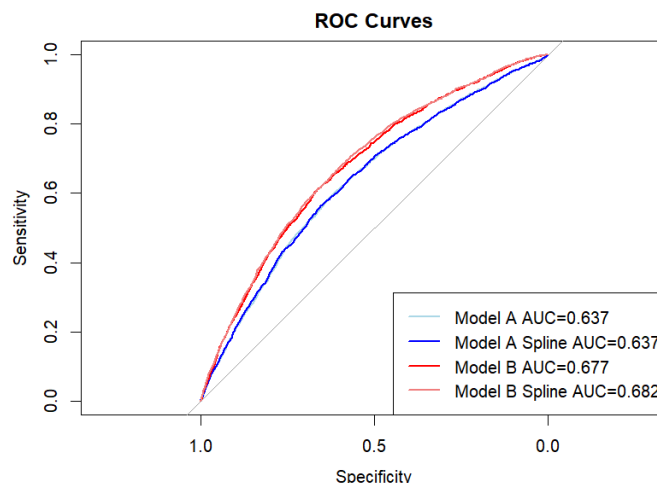
To determine the effectiveness of the models, we evaluate the G statistic and the mathematical calibration through the Hosmer-Lemeshow test.

- G-Statistic: Adding contextual control variables significantly improved the model. The G-Statistic doubled from 919.9815 to 1896.052. Introducing splines increased the value to 2043.076.

- Calibration (Hosmer-Lemeshow): the linearity on continuous variables forced rigid assumptions about the data, causing miscalibration, the Model A without splines had a total discrepancy of 18737.24, that means that the model's predicted probabilities are drastically different from what is in the data. By allowing non linearity in continuous variables using splines we drastically reduced the error to 42.41212. Nevertheless, the test fails as the p value should make us reject the null hypothesis (lower than 0.05). However, with 25313 observations, even small differences between the expected and the observed become detectable.

The AUC and ROC

The final Model B model with non linearity has achieved an AUC of 0.6818367, which means that if we randomly select a viral article and a non viral one, the model will assign a higher predicted probability to the viral article 68% of the time. For predicting human behaviour on the internet, obtaining an AUC near 0.7 proves that our model has a genuine sorting power. In this situation, the main improvement occurs when we add control variables to the model a with the improvement from a 0.6365 AUC to a 0.6768 in the model b. Nevertheless, slight improvements were as well made when we used splines in continuous variables with a 0.6369 AUC in the model A with splines and a 0.6818367 in the model B with splines. This is clearly reflected in the following plot:



Sensitivity and the issue with Accuracy and Specificity

As we established that only the top 25% of articles are considered viral, relying on accuracy would be misleading. For example, a model predicting 0 viral articles would have a 75% accuracy and a 100% specificity, because it identifies every non viral article. However, it would have 0% sensitivity. That is why our evaluation is focused on Sensitivity (True positive rate). While Accuracy remains stable across models, the transition from model A to model B with splines has a massive improvement in Sensitivity, increasing from 53.91% to 63.29%. This represents a 9-10% improvement in the model's ability to successfully identify true viral articles.

Statistical Discoveries

By extracting the Odds Ratios, we can draw conclusions on whether media and keywords drive virality or not. In this analysis we focus primarily on models A and B, specially when studying keywords, due to the fact that when we applied the Natural Cubic Spline we divided

the kw_avg_avg predictor into 3 mathematical segments. That is why we receive 3 keyword odds ratios in the result of the models in which we allow non linearity.

The multimedia Impact

One of the most interesting discoveries of this analysis is the statistical illusion induced by multimedia elements. In the first model, Model A, images appeared to have a strong relationship with the virality of articles. Images OR was 1.105 and the p value $2.773553e-12$. However, this result was conditioned by the Omitted Variables. When we studied the model B with splines, the Odds Ratios of images was $1.018139e+00$ and of videos $1.016615e+00$. The p value of both predictors as well increased significantly, the p value of images is $3.390663e-01$ and of videos $2.746470e-01$, this made these variables lose all statistical significance.

The Keywords Impact

Unlike multimedia, the strength of keywords (kw_avg_avg) survived the addition of control variables, proving to be a fundamental pillar of virality. As the continuous data was standardized prior to modeling, the Odds Ratios provide an interpretable metric. Interpreting the Odds Ratios of the Linear Model B, in constant conditions, for every 1 standard deviation improvement in keyword quality, the odds of an article becoming viral increase by 33.34%. The final p value for the kw_avg_avg predictor is $1.888237e-54$, which is essentially zero.

Conclusion

Virality is not achieved through multimedia elements. Once we mathematically control the actual quality and the context of the text, a mediocre article with images or videos is unlikely to become viral. However, through an excellent utilization of keywords we could help an article become viral. The final non linear logistic model with control variables is a context aware model with strong discriminatory power.

For this whole analysis, we have to understand that the virality of an article highly depends on human behaviour, which is impossible to predict perfectly. That is why we still consider the model to be reliable even though the test results are not perfect.